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12 September 1975

MILITARY SPECIFICATION

CATHETER AND NEEDLE UNIT, INTRAVENOUS, RADIOPAQUE, DISPOSABLE

(CATHETER INTO NEEDLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sterile, nontoxic, pyrogen-free, disposable, radiopaque, closed system intravenous placement units, suitable for prolonged intravenous therapy from a single venipuncture.

1.2 Classification. Catheter and needle units shall be furnished in the following types, classes, sizes and styles, as indicated below (see 6.1).

Types

I - Without 3-place connector
II - With 3-place connector

Class - Usable catheter length

1 - 12 inches
2 - 21 inches

Size - Catheter gage

A - 15
B - 18

Style - Needle gage

a - 14
b - 17

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-S-36586	- Sterilization Strip Test, Bacterial Spore.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

24367 Catheter and Needle Unit, Intravenous, Radiopaque, Closed System.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

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THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Bureau of Biologics. Food and Drug Administration Code of Federal Regulation Part 273, Minimum Requirements (as applicable).

(Application for copies should be addressed to the Food and Drug Administration, 5600 Fishers Lane, Rockville, MD 20852.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z70.1 American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the Americal National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Plastic and rubber components. All plastic and rubber components coming in contact with intravenous medication and patient's blood shall pass the test for extractable metals when tested as specified in 4.4.1. The total quantity of barium, cadmium and lead shall not exceed 5 parts per million (ppm) when tested as specified in 4.4.1. In addition, all plastic components coming in contact with intravenous medication or patient's blood shall meet the requirements of the USP for Class IV plastics, when tested as specified in 4.4.2.

3.1.1 Catheter. The catheter shall be fabricated from clear, pliant polyvinylchloride of the best quality and suitable in all respects for the purpose intended. The catheter shall be radiopaque throughout its length. The finished catheter shall be sterile, pyrogen free and shall pass the animal implantation test specified in 4.4.2.1.

3.1.2 Needle. The needle cannula shall be fabricated from seamless drawn, or welded and drawn tubing. Tubing shall be in accordance with GG-I-526, for class 3, type 304 material, and shall be subjected to the boil test as specified in GG-I-526.

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3.1.3 Extension tube. Extension tube shall be fabricated from clear, pliant, polyvinylchloride and shall meet the requirements specified in 3.1.

3.1.4 Extension tube hub and 3-place connector. The extension tube hub (applicable for both Type I and Type II units) and the 3-place connector (applicable for Type II units) shall be fabricated from a suitable rigid plastic and shall meet the requirements specified in 3.1.

* 3.1.5 Protective conduit. The protective conduit shall be fabricated from clear or translucent polyethylene.

* 3.1.6 Actuator and needle hub. Actuator and needle hub shall be fabricated from rigid or semi-rigid plastic. The needle hub shall meet the requirements specified in 3.1.

3.1.7 Needle guard, needle cover and protective caps. The needle guard, needle cover and protective caps shall be fabricated from a suitable plastic.

3.1.8 Injection sites (Type II design). The injection sites of the 3-place connector for units of Type II design shall be fabricated from natural latex rubber and shall meet the requirements specified in 3.1.

3.2 Style, design and dimensions. The catheter and needle intravenous unit shall be a sterile, nontoxic, pyrogen-free, disposable, radiopaque catheter placement unit, of the closed system design, suitable for prolonged intravenous therapy from a single venipuncture. Unit shall be ready for use and shall permit aseptic insertion by the surgeon without assistance, and without the necessity of hand scrubbing or wearing of sterile gloves. The unit shall consist essentially of an outer metal needle, of the short bevel design, an indwelling catheter that passes through the lumen of the needle, an extension tube with Luer hub, a protective conduit, an actuator to advance the catheter, an inline needle guard that locks the needle assembly, a needle cover that is discarded prior to venipuncture, and a protective cap for the exposed Luer cavity.

The needle assembly shall be enclosed into an inline needle guard. The needle guard shall permanently lock the needle hub after catheter placement in order to manipulate the indwelling radiopaque catheter after catheter insertion. The indwelling catheter shall form a safe, free sliding fit through the needle lumen. The inline needle guard shall completely encase the needle to prevent the needle bevel from shearing or piercing the catheter when the needle hub is locked into the restricting notches of the needle guard. The indwelling catheter shall be permanently secured to an extension tube. The extension tube shall have a larger internal diameter than the outside diameter of the catheter. The extension tube and catheter shall be permanently attached to the actuator. The actuator shall serve as a handle

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to physically advance the catheter into the vein by retaining the catheter in the sterile protective conduit without exposing the catheter. The protective conduit shall possess a longitudinal fine slit throughout its full length (the slit shall be self-closing and self-sealing) to effect easy advancement of the catheter without binding, kinking and physically touching the catheter when intravenous medication solution is allowed to flow through the system via the extension tube hub (or 3-place connector for Type II units) prior to venipuncture. This design is defined to be the closed system technique utilizing intravenous solution as the stylet. The protective conduit shall be securely retained into the needle hub. The actuator shall be designed to snap into the needle hub of the needle guard assembly to effect a secured locked junction while simultaneously releasing the protective conduit. Attached to the proximal end of the extension tube shall be a female Luer hub. The Luer hub shall possess a cavity in accordance with Figure 2, female end, of ANSI Standard Z70.1, and in addition shall meet the requirements specified in 3.2.1. All units shall be furnished with a needle cover that shall be discarded prior to venipuncture. Type II units shall have a 3-place connector attached to the extension tube hub. The male port of the 3-place connector shall possess an external Luer taper in accordance with Figure 2, male end of ANSI Standard Z70.1, and in addition shall meet the requirements specified in 3.2.2. The top center port of the connector shall be a Luer hub possessing a female Luer cavity in accordance with Figure 2, female end, of ANSI Standard Z70.1, and in addition shall meet the requirements specified in 3.2.1. The two remaining top ports of the 3-place connector shall be injection sites for additional medication infusion. These ports shall be fabricated from rubber and shall be securely attached to the arms of the connector. The rubber ports shall completely reseal upon puncture with needle gages up to and including 19 gage. Type I units shall possess a protective cap covering the Luer hub of the extension tube. Type II units shall possess a protective cap covering the center portion of the 3-place connector. Type I and Type II catheter and needle units shall be in accordance with DPSC Drawing No. 24367 for applicable size, style, stock number and item identification.

3.2.1 Diameter and depth of Luer hub cavities. The female Luer hub cavity of the extension tube and 3-place connector shall be tested for diameter and depth of Luer hub cavity as specified in 4.4.3. Hubs which do not fit the plug gage within the "GO - "NO GO" limits specified in 4.4.3, shall be rejected. In addition, when the hubs are fitted onto the plug gage, there shall be no movement of the hubs on the gage when the hub is pressed lightly from side to side in all directions.

3.2.2 Male port of 3-place connector. The male Luer taper of the 3-place connector shall be tested for proper taper as specified in 4.4.4. Male Luer tapers that do not fit the ring gage within the "GO - "NO GO" limits specified in 4.4.4, shall be rejected. In addition, when the ring gage is fitted on the male taper of the connector, there shall be no movement of the gage on the connector when the gage is pressed ligntly from side to side in all directions.

3.3 Security of assembly.

3.3.1 Catheter to extension tube and actuator. When tested as specified in 4.4.7.1, the extension tube and catheter shall not break, loosen, or pull out of the actuator; further the catheter shall not separate from the extension tube.

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3.3.2 Extension tube to extension tube hub. When tested as specified in 4.4.7.2, the extension tube shall not break, or loosen from the extension tube hub.

3.3.3 Protective conduit to needle hub. When tested as specified in 4.4.7.3, the protective conduit shall not separate from the needle hub.

3.3.4 Needle to needle hub. When tested as specified in 4.4.7.4, the needle shall not separate from the needle hub.

3.4 Flexibility of catheter. When tested as specified in 4.4.8, the plastic catheter shall not crack or collapse. In addition, liquid flow shall not be obstructed in the lumen of the catheter when tested as specified in 4.4.8.

3.5 X-ray opacity. When tested as specified in 4.4.11, the plastic catheter shall appear prominent and conspicuous when covered by at least 5/8 inch thickness of aluminum.

3.6 Resistance to breakage of the needle cannula. When tested as specified in 4.4.9, the needle cannula shall not break.

3.7 Leakage of catheter and extension tube. When tested as specified in 4.4.10, the junction of the extension tube to the catheter shall not show leakage; the connection of the extension tube to the extension tube hub shall not show leakage; and the Luer connection of the extension tube hub shall not show leakage exceeding the formation of one clinging drop. As an alternate test, the above connections shall not show leakage exceeding the formation of one clinging air bubble when tested as specified in 4.4.10.1. For type II units, in addition to all of the above, the male Luer connection of the 3-place connector to the extension tube hub and the female Luer top port of the 3-place connector shall not show leakage as specified above.

3.8 Freedom from pyrogens. All components of the unit shall be free from pyrogens when tested as specified in 4.4.6.

3.9 Sterility. All components of the unit shall be sterile when tested with a biological indicator as specified in 4.4.5.

3.9.1 Certificates of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

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a. Item identification.
b. National stock number (NSN).
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form a biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa. 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.9.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simutaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.10 Animal implantation test. The plastic catheter member only shall be tested, and shall show no evidence of toxicity or adverse tissue reaction upon gross and microscopic examination, when tested as specified in 4.4.2.1.

3.11 Instructions. A detailed instruction sheet for proper use of the catheter and needle units shall be provided in each immediate container. A caution note in the instructions shall state words to the effect:

"DO NOT ATTEMPT TO RETRACT CATHETER WHILE THE NEEDLE IS IN THE VEIN".

3.12 Certificates of freedom from pyrogens, biological tests, and animal implantation test. A certificate listing all lot or control numbers shall state that representative samples from each lot or control have been tested in accordance with methods described in the USP and found to be free from pyrogens,

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and have passed the biological tests and the animal implantation test. This certificate shall include the specific test performed and the test results. The record of the animal implantation test and its results shall be certified by the pathologist who performed or supervised performance of the animal implantation test. These certificates shall be submitted by the contractor in the manner as specified in 3.9.1.

3.13 Manufacturer's record. The manufacturer shall place a code or lot number on the immediate container which can be verified against the manufacturer's records of all tests made on sterile, pyrogen-free units. These records shall be made available for inspection by any duly qualified official of the Federal Food and Drug Administration. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.14 Compatibility. The materials of which the catheter and needle units are manufactured shall not react with, or be acted upon, by the liquids with which they will be in contact during normal use and operation nor shall there be any material incorporated in the units which will cause any deleterious effect when the units are used for the purpose intended.

3.15 Workmanship. Workmanship shall be first class throughout. The finished items shall be free from defects which detract from their appearance or impair their serviceability. The indwelling catheter, needle cannula and plastic extension tube shall show a smooth outer surface, free from pits, tool marks, burrs, scale, corrosion, and molding flash. Inside surfaces of all components of the unit shall be free from scale, deep pits, deep furrows and die marks. The lumen of the catheter, needle cannula, hub cavities, protective conduit and actuator shall be free from burrs, chips, dirt, slivers, imbedded particles, or other foreign matter. Needles shall be sharp, not blunt or bent, and points shall be free from feathered edges, burrs or other defects, when examined under 10 power (10X) magnification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the polyvinylchloride, polyethylene, corrosion-resistant steel, and rubber (for Type II units) may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one complete catheter and needle unit.

TABLE I. Sampling criteria for examination.

Type defect	Inspection level	AQL (% defective)
Critical	III	1.0
Major	II	1.0

4.2.2 For Tests.

4.2.2.1 For sterility test. Sampling for sterility test shall be in accordance with the USP for parenteral devices and as specified in 4.4.5.

4.2.2.2 For pyrogen test. Sampling for pyrogen test shall be as indicated in the USP for transfusion and infusion assemblies. Samples shall be selected which are to be representative of each lot of plastic used in the fabrication of the units. As an alternate, sampling shall be on a sterilizer lot basis when sterilizer lots are smaller than, or equal to, production lots.

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* 4.2.2.3 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II. The unit of product for testing purposes shall be one complete catheter and needle unit. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling criteria for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (% defective)
	Composition analysis	3.1.2 & GG-I-526	GG-I-526	1/	--
	Boil test	3.1.2 & GG-I-526	GG-I-526	S-1	1.0
Needle Cannula	Dimensions	3.2	--	S-1	2.5
	Security of assembly	3.3.4	4.4.7.4	S-1	0.65
	Resistance to breakage	3.6	4.4.9	S-1	1.0
Plastic	Extractable metals	3.1	4.4.1	2/	--
Catheter	Material-Class IV	3.1	4.4.2	2/	--
	Dimensions	3.2	--	S-1	2.5
	Flexibility test.	3.4	4.4.8	S-1	1.0
	X-ray opacity	3.5	4.4.11	S-1	1.0
	Animal implantation test	3.10	4.4.2.1	-	-
Extension tube and	Extractable metals	3.1	4.4.1	2/	--
	Material-Class IV	3.1	4.4.2	2/	--
Extension tube hub	Dimensions	3.2	--	S-1	2.5
	Diameter and depth of Luer hub cavity	3.2.1	4.4.3	S-1	0.65
	Security of assembly	3.3.2	4.4.7.2	S-1	0.65
Plastic Catheter and Extension Tube	Security of assembly	3.3.1	4.4.7.1	S-1	0.65
	Leakage test	3.7	4.4.10 or 4.4.10.1	S-1	0.65

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TABLE II. Sampling criteria for tests. (cont'd)

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (% defective)
Protective Conduit	Security of assembly	3.3.3	4.4.7.3	S-1	0.65
3-place Connector (Type II unit only)	Extractable metals	3.1	4.4.1	2/	--
	Material-Class IV	3.1	4.4.2	2/	--
	Diameter and depth of Luer cavity	3.2.1	4.4.3	S-1	0.65
	Male Luer taper	3.2.2	4.4.4	S-1	0.65
Needle hub	Extractable metals	3.1	4.4.1	1/	--
	Material-Class IV	3.1	4.4.2	2/	--

1/ Sufficient quantity of each lot of metal used in the construction of the needles shall be selected for chemical analysis.
2/ Sufficient samples to perform the extractable metal tests and biological tests shall be selected from each lot and shall be representative of each batch of plastic and rubber (injection sites - Type II units) used in the manufacture of the catheter and needle units.

4.3 Examination.

* 4.3.1. Examination of catheter and needle units. The complete catheter and needle units shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any units in the sample containing one or more defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

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TABLE III. Classification of defects.

Categories	Defects
Critical	
01	Catheter and extension tube not permanently secured to each other and to actuator.
02	Catheter pierced by needle.
03	Complete unit not free from foreign matter.
Major	
101	Catheter and needle units not of the proper type, class, size and style.
102	One or more components missing.
103	Catheter surface not free from kinks, dents, pits, blisters, cuts, holes and scratches.
104	Needle point not keen, not straight, and not free from feathered edges.
105	Needle surfaces not free from dents, pits, burrs, slivers, scale and corrosion.
106	Catheter does not form a safe, free sliding fit through the needle lumen when the actuator is advanced toward the needle guard.
107	Protective conduit separates from needle hub before actuator is locked into place.
108	Longitudinal slit of protective conduit not self-sealing and self-closing.
109	Needle guard does not completely encase the needle bevel and does not completely lock needle hub when the needle hub is fully engaged into restricting notches.
110	Extension tube hub not permanently secured to extension tube.
111	Needle cannula not permanently secured to needle hub.
112	Fluid inlets not covered by protective caps.
113	Rubber injection sites of 3-place connector for Type II units do not reseal when punctured by a size 19 gage needle.
114	Protective conduit opaque.
115	Instructions for use and caution note missing.

* 4.3.2 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

* 4.4.1 Extractable metals. All plastic components (excluding needle covers, needle guard, protective caps, protective conduit, actuator, and needle hub collar) and the rubber injection sites of the 3-place connector (for Type II units) shall be tested for extractable metals. The specimens shall be cut such that no thickness exceeds 3 millimeters (mm) and eluted for either 72 hours at 50 degrees Centrigrade (^{o}C), 24 hours at 70° C or 60 minutes at 121° C in two menstrua: 0.9 percent saline and cotton seed oil. Each gram of each specimen shall be covered with 5 milliliters (ml) of the menstruum (recommended amount is 10 grams of the specimen plus 50 ml of solvent). The menstrua shall be collected, after agitation, at the end of the selected elution period and submitted to spectographic analysis for barium, cadmium and lead. The metals shall be reported in ppm in the elutate. A value of 1.0 ppm in the eluate shall be equivalent to extracting 1.0 milligram from 200 grams of test material in the elution period into 1000 ml elution menstruum.

4.4.2 Biological tests for Class IV plastics. All plastic components coming in contact with intravenous medication and patient's blood shall be tested in accordance with the requirements of the USP for Class IV plastics.

4.4.2.1 Animal implantation test. Samples of the finished indwelling plastic catheter (end item) shall be tested in accordance with the USP for Implantation Test, Plastic Containers, for Class VI plastics, except after "72 hours" add "and not to exceed 144 hours"; and delete 'Examine macroscopically the area of the tissue surrounding the center portion of each implant strip. Use a magnifying lens if necessary,' and substitute: "The muscle tissue surrounding the implants shall be examined, both grossly and microscopically, by a pathologist certified by the American Board of Pathology or the American College of Veterinary Pathologists'."

4.4.3 Diameter and depth of Luer hub cavities. The Luer hub cavity of the extension tube (Types I and II units) and the 3-place connector (Type II units) shall be tested with male Luer taper plug gage in accordance with Figure 5 of ANSI Standard Z70.1 (this gage shall be as shown in the ANSI Standard, except that it shall be constructed with a suitable liquid or pneumatic path to permit performance of the leakage test specified in 4.4.10). The female Luer cavities shall be slipped onto this gage with a straight motion, without turning or twisting the plug gage into the cavity. Once the female Luer cavity is slipped onto the gage, no additional force shall be exerted to force the plug gage into the Luer cavity. The fit of the Luer cavities onto this gage shall be tested by means of the feeler gage in accordance with Figures 5 and 6 of ANSI Standard Z70.1.

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4.4.4 Male port of 3-place connector. The external taper of the male port of the 3-place connector (type II units only) shall be tested with the female Luer ring gage in accordance with Figure 7 of ANSI Standard Z70.1. The male taper shall be slipped into this gage with a straight motion, without turning or twisting the male taper into the gage. Once the male taper is slipped into the gage, no additional force shall be exerted to force the male taper of the connector into the gage.

4.4.5 Sterility test. All components of the catheter and needle units shall be tested for sterility in accordance with the method specified in the USP for parenteral devices using innoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carier). The incubation temperature shall be as indicated in paragraph 4.4.5.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product tests units.

4.4.5.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C
Dry Heat	B. stearothermophilus, or	55° - 65°C
	B. subtilis var. niger (or globigii)	30° - 35°C
Gas	B. subtilis var. niger (or globigii)	30° - 35°C
Radiation	B. pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.5.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of tests units shall be as indicated in 4.4.5 Sterility test and 4.4.5.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.5.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

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4.4.5.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method.

4.4.5.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

4.4.6 Freedom from pyrogens. Samples of the finished units shall be tested for freedom from pyrogens in accordance with the USP for transfusion and infusion assemblies.

4.4.7 Security of assembly tests.

4.4.7.1 Catheter to extension tube and actuator. The security of assembly of the plastic catheter to the extension tube and the catheter-extension tube bond to the actuator shall be tested by exerting a steady dead load of 3 pounds for a period of 5 minutes, with the actuator appropriately secured. The load shall be applied to a segment of the extension tube, with the extension tube hanging vertically downward.

4.4.7.2 Extension tube to extension tube hub. The security of assembly of the extension tube junction to the extension tube hub shall be tested by exerting a steady load of 6 pounds for a period of 5 minutes, with the hub appropriately secured. The load shall be applied to a segment of the extension tube, with the extension tube hanging vertically downward.

4.4.7.3 Protective conduit to needle hub. The security of assembly of the protective conduit to the needle hub shall be tested by exerting a steady load of 3 pounds for a period of 3 minutes, with the needle hub appropriately secured. The load shall be applied to the proximal end of the conduit with the conduit hanging vertically downward and the actuator placed against the most adjacent stop of the conduit.

4.4.7.4 Needle to needle hub. The security of assembly of the needle cannula junction to the needle hub shall be tested by exerting a steady load of 10 pounds for a period of 5 minutes, with the needle hub appropriately secured.

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4.4.8 Catheter flexibility test. The plastic catheter shall be wrapped 360 degrees around a circular rod or mandrel, 3/8 inch in diameter. The condition of the catheter shall be noted when released from the rod or mandrel. A suitable liquid shall then be allowed to flow through the lumen of the catheter within 5 seconds under an internal pressure of 3 ± 1/2 pounds per square inch gage (psig). The liquid shall be allowed to enter the catheter at the extension tube hub end.

4.4.9 Resistance to breakage. The hub of the needle cannula shall be clamped in a fixed position. The needle cannula shall then be bent from its original neutral position in a clockwise direction through an arc of 5 to 10 degrees, by applying a force to the cannula at a contact distance of 1-1/4 inches from the needle hub. A force shall then be applied in a counter-clockwise direction (without changing contact distance) to bend the cannula past its neutral position to an angle of 5 to 10 degrees. The cannula shall then be returned to its neutral position.

4.4.10 Leakage test (hydrostatic method). With the extension tube in a horizontal or vertical position, the extension tube hub (for Type II units, the female hub of the 3-place connector) shall be inserted onto the plug gage specified in 4.4.3 to a water pressure source having a pressure gage. The female hub shall be inserted onto the plug gage with a straight motion, without turning or twisting the plug gage into the hub cavity with a force not to exceed 5 pounds. The feeler gage, specified in 4.4.3, shall be placed onto the plug gage to assure that the "GO" position is not violated. Water shall be admitted to flow through the lumen of the extension tube and catheter so as to fill the cavity space and the lumen of the extension tube and catheter. The catheter shall then be sealed at its distal end in a leakproof manner, and a pressure of 45 psig shall be applied for at least 10 minutes.

4.4.10.1 Leakage test (pneumatic method). This test shall be performed similarly to the test specified in 4.4.10, except that air at 35 psig shall be the testing media. The complete assembly of the item to connected plug gage shall then be completely immersed into a suitable container of water, with the air pressure applied for at least 6 seconds. The entire unit shall be observed for leakage as evidenced by air bubbles.

4.4.11 X-Ray Opacity test. The plastic indwelling catheter member only shall be placed on an unexposed X-ray plate and covered with an aluminum ladder ranging in thickness from at least 1/8 inch to 3/4 inch, in steps of 1/8 inch. X-ray exposure shall be as listed below; accomplished with automatic processing and using appropriate speed screens in a convenient size cassette:

Milliampere-seconds	(MAS)	----------	10
Peak kilovolts	(KVP)	----------	65 to 70
Focal length distance (inches)		----------	30 to 40

The above stated exposure shall clearly delineate the steps 1/4 inch, 3/8 inch, 1/2 inch and 5/8 inch from each other.

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5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package containing ten catheter and needle units, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Individual package. Each catheter and needle unit shall be packaged in a sealed peel open type bag capable of maintaining sterility of contents unless the container is damaged or opened. The container shall be designed with tabs to facilitate opening. The unit shall be so packaged to prevent the tubing from being kinked or distorted.

5.1.2.2 Unit package. Ten individual packages containing catheter and needle units of one type, class, size and style shall be packaged in a snug fitting box constructed in accordance with PPP-B-636, class weather-resistant. Closure and waterproofing shall be as specified in the appendix of the box specification.

5.1.3 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Five intermediate packages containing Type I, catheters of one size only or ten intermediate packages containing Type II catheters shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B. Five intermediate packages containing Type I catheters of one size only or ten intermediate packages containing Type II catheters shall be packed in an exterior container designed for a Type II load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1 or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in the appendix of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Individual package. In addition to any commercial markings, each individual package shall be marked with the lot (control) number, date of manufacture, and the requirements of the Federal Food, Drug and Cosmetic Act. Marking shall include the following legend or similar statement: "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED".

5.3.2 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Lot (control) number, contract or purchase order and name of contractor shall be shown. The method of sterilization (G-Gas, R-Radiation, S-Steam, C-Cold, D-Dry Heat Sterilization) shall be shown following the level of packaging and date of packaging. Example: "A-5/75-G". Final sterilization shall be on or after indicated packaging date.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number and method of sterilization as specified in 5.3.2 shall be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class, size and style required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Applicable National Stock Number.

6.2 National stock number coverage. This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification	Specification Designation			
		Type	Class	Size	Style
6515-00-138-2298	CATHETER AND NEEDLE UNIT, INTRAVENOUS, 15 Gage Catheter, Radiopaque, Disposable, 12 inches Long, 10s	I	1	A	a
6515-00-138-2299	CATHETER AND NEEDLE UNIT, INTRAVENOUS, 18 Gage Catheter, Radiopaque, Disposable, 12 inches Long, 10s	I	1	B	b
6515-00-138-2307	CATHETER AND NEEDLE UNIT, INTRAVENOUS, 15 Gage Catheter, Radiopaque, Disposable, 21 inches Long, 10s	II	2	A	a

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* 6.3 Changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and Contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
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Navy - MS
Air Force - 03

Preparing Activity:
Defense Logistics Agency - DM

Project No. 6515-2068

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-37203A CATHETER AND NEEDLE UNIT, INTRAVENOUS, RADIOPAQUE, DISPOSABLE (CATHETER INTO NEEDLE)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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